(From Trivedi)

Parameter Estimation

Confidence Interval

- Point Estimate $\hat{\theta}$ rarely coincides with the actual value
- Interval Estimate – Desirable ✓
- The estimator satisfies (1)

$$P(\hat{\theta} - \epsilon_1 < \theta < \hat{\theta} + \epsilon_2) = \gamma$$

then the random interval $A(\theta) = (\hat{\theta} - \epsilon_1, \hat{\theta} + \epsilon_2)$ is $100\gamma\%$ confidence interval for θ.

γ – confidence coefficient

Apply Chebyshev's inequality;

$$P(\hat{\theta} - \epsilon < \theta < \hat{\theta} + \epsilon) \geq 1 - \frac{var[\hat{\theta}]}{\epsilon^2} \quad (2)$$

provided $var[\hat{\theta}]$ is known or can be estimated.

Ex Let $\theta = \mu$ and $\hat{\theta} = \bar{x}$

Assume σ^2 is known (σ^2 is popn. variance)

Therefore $P(\bar{x} - \epsilon < \mu < \bar{x} + \epsilon) \geq 1 - \frac{\sigma^2}{n\epsilon^2}$ (3)

For large sample size $n \to \infty$, we can make confidence coefficient close to 1.

Sampling from Normal Distribution

Random sample of size n is taken from a normal population with <u>unknown mean</u> and <u>Known</u> variance.

$$X \sim N(\mu, \sigma^2) \qquad (4)$$

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ so that } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

To find a and b such that (5)

$$P(a < Z < b) = \gamma.$$

Therefore the required confidence interval (6)

$$a < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < b \qquad (7)$$

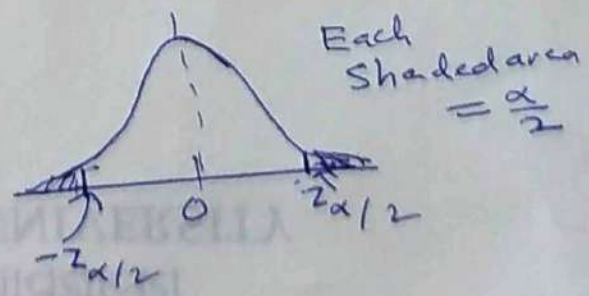
or $$\boxed{\bar{x} - \frac{b\sigma}{\sqrt{n}} < \mu < \bar{x} - \frac{a\sigma}{\sqrt{n}}}$$

Therefore $\left(\bar{x} - \frac{b\sigma}{\sqrt{n}}, \bar{x} - \frac{a\sigma}{\sqrt{n}}\right)$ is a $100\gamma\%$ confidence interval for μ.

Ex If $a = -b$, then

$$P(-b < Z < b) = \gamma$$

Let $\gamma = 1 - \alpha$, then from symmetry of pdf of Z, $P(Z < -b) = \frac{\alpha}{2}$ and $P(Z > b) = \frac{\alpha}{2}$ (8)



$100(1-\alpha)\% = 100\gamma\%$ CI for mean μ.

(9)

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

<u>Note</u>: In practice assumption of normality does not pose problem

– Central limit theorem states that the statistic $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ asymptotically is normal.

<u>Problem</u> If σ^2 is unknown
Replace σ^2 by estimate s^2

We have (10)

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

has student t-distribution with $n-1$ degrees of freedom.

Rohatgi/Saleh

<u>Chi-Square χ^2, t-Distributions : Exact Sampling Distributions</u>

Let $X_1, \ldots X_n$ be iid rvs and let $S_n = \sum X_i$. Then

(a) $S_n \sim \chi^2_n \iff X_1 \sim \chi^2(1)$

(b) $X_1 \sim N(0,1) \Rightarrow \sum_{k=1}^{n} X_k^2 \sim \chi^2_n$

If X has a χ^2 chi-square distn. with n df we write $X \sim \chi^2_n$

$$\boxed{\text{Special Case of Gamma distn.}} \quad f(x) = \begin{cases} \dfrac{x^{n/2-1} e^{-x/2}}{2^{n/2}\,\Gamma(n/2)} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

t - distribution

Let $X \sim N(0,1)$ and $Y \sim \chi^2_n$ and let X and Y be independent

$$T = \frac{X}{\sqrt{Y/n}} \sim t_n$$

PDF of T is

$$f_n(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)\sqrt{n\pi}} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}, \quad -\infty < t < \infty$$

* [$n = 1$ is Cauchy rv

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SESSION – XII (B) p5–11

(Trivedi) HYPOTHESIS TESTING

Statistical Inference
Main Problems

Estimation of Parameters | Tests of statistical hypothesis

Statistical Hypothesis – A statement or assertion about one or more probability distribution associated with population

Statistical Tests Procedures to decide whether to reject or accept hypothesis based on information contained in a sample. $x_1, x_2, \ldots, x_n$ from a population distribution F_θ.

↓

Functional form is known except parameter θ.

Examples

1. Coin Tossing – whether prob. p of head equals p_0.
2. System Administrator – to decide to upgrade the capacity of the installation.
3. Battery life – T hours (lasts)

We need to take a random sample to test the hypothesis

We form <u>Null hypothesis</u> (claim about prob. distn)

<u>Alternative hypothesis</u> (Contradicting hypothesis)

H_0 (Null hypothesis) vs H_1 (Alternative hypothesis)

<u>Example 1</u>: MTTF (mean time to failure)

H_0: MTTF exceeds say θ_0 hours

H_1: MTTF $< \theta_0$

<u>Example 2</u> Manufacturing process of certain electronic components. have average life μ hrs

-> It is proposed to test a new procedure for manufacturing of electronic component!

H_0: $\mu = \mu_0$

H_1: $\mu > \mu_0$ or $\mu < \mu_0$ or $\mu \neq \mu_0$

From Rohatgi & Saleh) <u>Defn</u> <u>Parametric Hypothesis</u>

— is an assertion about the unknown parameter θ. It is referred to as <u>Null hypothesis</u> $H_0: \theta \in \Theta_0 \subset \Theta$,

— the statement $H_1: \theta \in \Theta_1 = \Theta - \Theta_0$ is usually referred to as alternative hypothesis H_1

Random sample $X_1, X_2, \ldots X_n$.

X_i is a rv for each i

R^n (: the totality of all n-tuples will span Euclidean n-space R^n).

Divide n-space R^n into two regions

(Acceptance Region) $R(H_0)$ $\qquad$ $R(H_1)$ (Critical or Rejection region)

<u>Decision Rule</u> : If sample point $\underset{\sim}{x} = (x_1, x_2, \ldots, x_n)$ lies in $R(H_1)$, we reject H_0

<u>On the other hand</u> If the observed n-tuple lies in $R(H_0)$, we fail* to reject H_0

* <u>Note</u>: The sample does not have enough evidence against H_0

<u>Two Types of Errors</u>

Type I error – can be made – may reject H_0 when true

Type II error – Accept H_0 when false

	True: H_0	True: H_1
H_0	Correct	Type II Error
Accept H_1	Type I error	Correct

$P(\text{Type I error}) = \alpha$

$\downarrow$

Ho True but sample lies in rejection region (leading us to reject Ho)

α is known as Level of significance of the test

$P(\text{Type II error}) = \beta$

Ho false but sample lies in Acceptance Region

$(1-\beta)$ is known as power of the test.

<u>Summary</u>

$$\alpha = P(\text{Type I error})$$
$$= P(\text{Rejecting Ho when it is true})$$

$$\beta = P(\text{Type II error})$$
$$= P(\text{Accepting Ho when it is False})$$

<u>Wrong Decision</u> : Type I or Type II errors lead to Wrong Decision

— Attempt to minimize these errors

<u>Fix sample size</u> : a decrease in one type error leads to an increase in other type of error.

1. If C is the critical region, then $P_\theta C$ $\theta \in \Theta_0$, (Prob of Type I error)

where $C = \{ \underset{\sim}{x} \in \mathbb{R}^n : H_0 \text{ rejected if } \underset{\sim}{x} \in C \}$

2. $P_\theta C^c$, $\theta \in \Theta_1$ is (Prob: Type II error)

Partition the sample space into two disjoint sets C and C^c

<u>Ideal Situation</u>: Find critical region for which these probabilities are zero.

— Such situations do not arise in practice

<u>Procedure in Practice</u>:

Rohatgi & Saleh

Limit Type I error to a prearssigned level α say (0.01 or 0.05)

Minimize prob. of Type II error.

Tests on the Population Mean

Simple Hypothesis $H_0 : \theta = \theta_0$ (parameter θ of the population)

<u>Steps</u>:

1. Find statistic $W = W(X_1, X_2, \cdots X_n)$ such that distribution of W is known.
2. Choose an interval (a, b) such that $P(W \notin (a,b) \mid H_0 \text{ is true}) = \alpha$

We can write

$$P(a < W < b \mid H_0 \text{ true}) = 1-\alpha$$

i.e.

$$P(a < W(x_1, \ldots x_n) < b) = 1-\alpha$$

3. Take a sample $x_1, \ldots x_n$ and compute $w = W(x_1, \ldots x_n)$
if $w \notin (a,b)$ reject H_0 in favour of H_1;
otherwise fail to reject H_0

<u>Problem</u> (Trivial) Test a hypothesis regarding population mean μ based on a random sample of size n from a normal population with known variance.

$$H_0 : \mu = \mu_0$$
$$H_1 : \mu \neq \mu_0$$

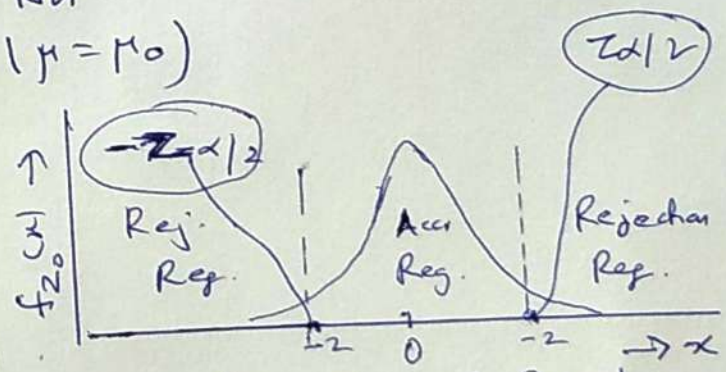
Test Statistic $\quad Z = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \qquad \underline{(1)}$

Note $Z \sim N(0,1)$.

Let $\quad Z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \qquad \underline{(2)}$

$$\alpha = 1 - P(-b < z < b \mid \mu = \mu_0)$$

$$b = z_{\alpha/2} \qquad \underline{(3)}$$



(Acceptance & Rejection Regions)

<u>Acceptance Region</u>

$$-z_{\alpha/2} < Z_0 < z_{\alpha/2} \qquad \underline{(4)}$$

or $\quad \mu_0 - \dfrac{z_{\alpha/2}\sigma}{\sqrt{n}} < \bar{X} < \mu_0 + \dfrac{z_{\alpha/2}\sigma}{\sqrt{n}}$.

Rejection region (5)

$$|\bar{x} - \mu_0| > \frac{z_{\alpha/2}\,\sigma}{\sqrt{n}}$$

Ex. (Trivedi)

Program's average working set size

$\mu_0 = 50$ pages $\sigma^2 = 900$ pages^2

Reorganization decreased working set size

Test the hypothesis

$H_0: \mu = \mu_0$ versus $H_1: \mu < \mu_0$

$Z_\alpha = Z_{0.05} = 1.645$ at 5% level of significance

The critical region

$$\bar{x} < \mu_0 - \frac{\sigma}{\sqrt{n}} Z_\alpha = 50 - \frac{30}{\sqrt{n}} \times 1.645$$

or $\bar{x} < 50 - \frac{49.35}{\sqrt{n}}$

Take samples $n = 100$

If $\bar{x} < 45$, we have reason to believe reorganization improved program locality